

Chapter 5B: Equations

4 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-EQU-C-01 solves linear equations of up to two steps.

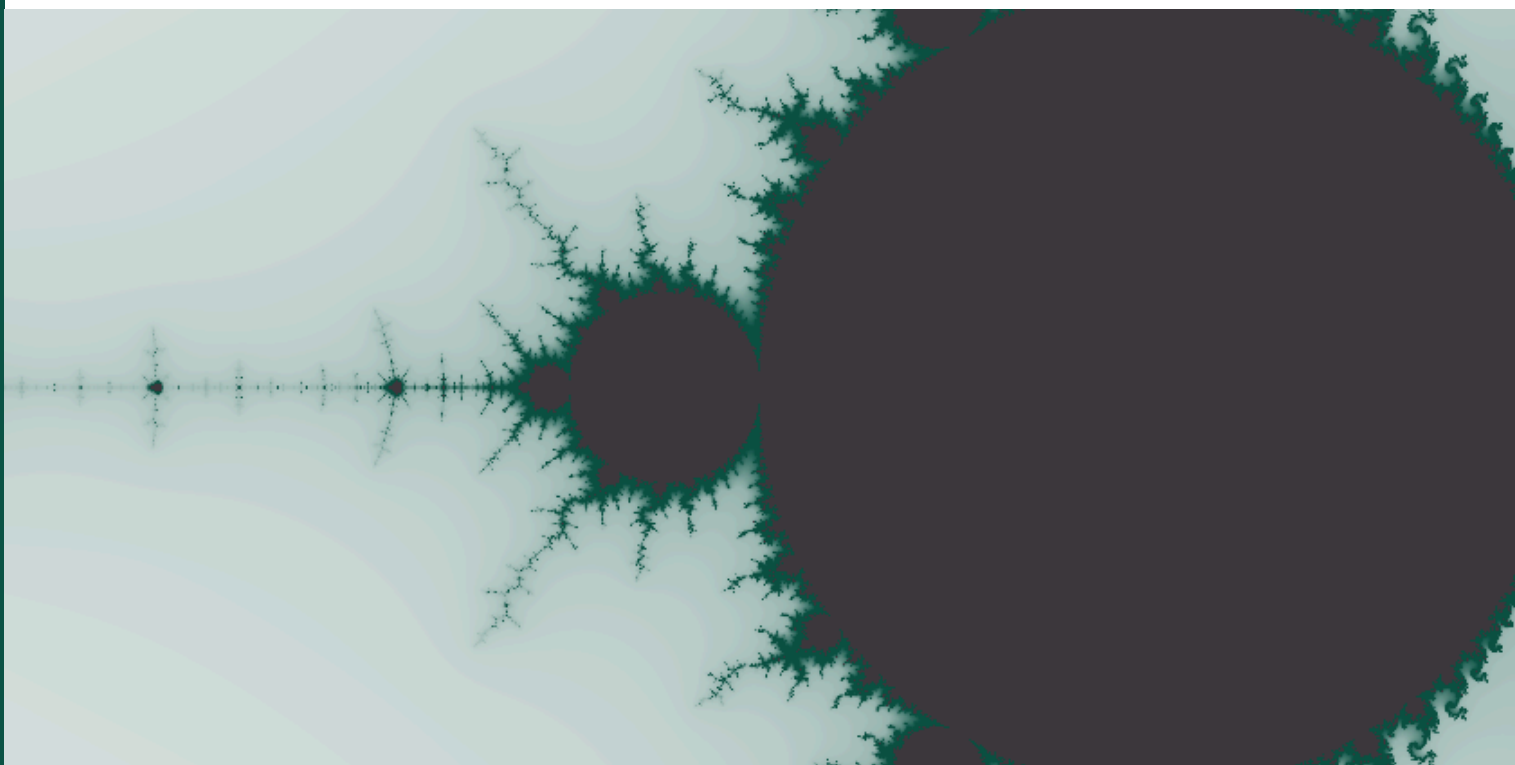


5.10 Guess, check and improve

5.11 Solving one-step equations

5.12 Solving two-step equations

5.13 Equations and problem solving



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

WE DO **Grey boxes.** Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

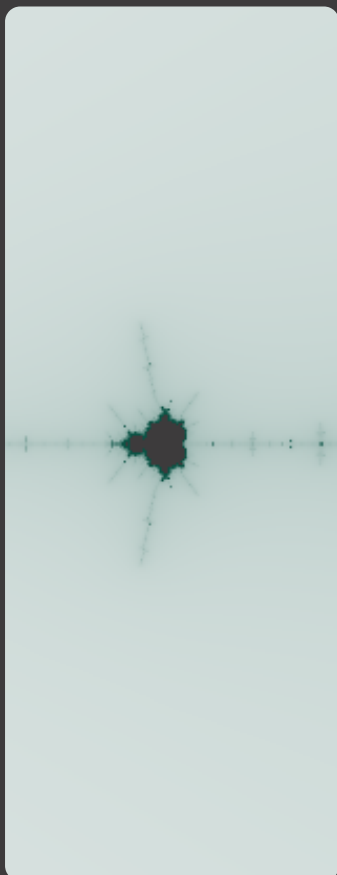
4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

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5.11	Solving one-step equations	7		☐
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In this chapter you will

- check whether a value is a solution
- solve equations by guess, check and improve
- solve one-step and two-step equations using inverse operations
- write and solve equations to solve problems



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
5.10

Guess, check and improve

Date _____

Learning intentions

We are learning to:

- solve equations by guess, check and improve

Success criteria

I can:

- ☐ substitute a guess into the equation
- ☐ decide whether the guess is too high or too low
- ☐ improve the guess until it is correct

Key terms Write the definition as your teacher explains it.

Equation

Solution

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Is the value a solution? Then solve by guess, check and improve.

a Is $x = 4$ a solution of $3x + 2 = 14$?

b Solve $4m - 9 = 7$.

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Is the value a solution? Write yes or no.

12 QUESTIONS

REMEMBER Guess a value for the variable.

1 $x + 5 = 12$, $x = 7$

2 $3y = 21$, $y = 6$

3 $m - 4 = 9$, $m = 13$

4 $2k + 1 = 11$, $k = 5$

5 $\frac{p}{2} = 8$, $p = 4$

6 $10 - t = 3$, $t = 7$

Questions 7–12: solve by guess and check.

7 $x + 7 = 15$

8 $3y = 24$

9 $m - 5 = 9$

10 $2k + 3 = 11$

11 $5n = 45$

12 $p - 12 = 3$

YOU DO

On your own: **Easy** Set B. Solve by guess, check and improve.

6 QUESTIONS

1 $x + 8 = 15$

4 $2a + 1 = 9$

2 $4m = 36$

5 $3n - 2 = 10$

3 $y - 6 = 11$

6 $\frac{w}{5} = 4$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Solve by guess, check and improve.

a Solve $\frac{k}{7} + 2 = 2$.

Your teacher works this out. Copy every step.

b Solve $\frac{d}{3} - 4 = 2$.

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Solve by guess, check and improve. Show your guesses in a table.

6 QUESTIONS

REMEMBER Substitute it into the equation.

1 $4m - 9 = 7$

3 $\frac{d}{3} - 4 = 2$

5 $7b - 10 = 39$

2 $\frac{k}{7} + 2 = 2$

4 $5a + 13 = 58$

6 $3(x + 2) = 27$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve by guess, check and improve.

a Solve $5a + 13 = 58$.

Your teacher works this out. Copy every step.

b Solve $2x + 3 = 4$ (the answer is not a whole number).

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Solve by guess, check and improve.

4 QUESTIONS

REMEMBER Too big or too small? Adjust your guess.

1 $2x + 3 = 4$

3 $x^2 + x = 30$

2 $n^2 = 144$

4 I think of a number, multiply by 6 and subtract 5. The answer is 43. What is my number?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Is $a = 3$ a solution of $5a - 1 = 14$?

Level 2

Solve $3x + 4 = 25$.

Level 3

Solve $\frac{y}{4} + 7 = 12$.

 Extension · Stretch yourself Worked example

Solve $x^2 = 20$ to 1 decimal place by guess, check and improve.

1. $4^2 = 16$ (low), $5^2 = 25$ (high)
2. $4.4^2 = 19.36$ (low), $4.5^2 = 20.25$ (high)
3. $4.47^2 \approx 19.98$, so round to 4.5

Answer: $x \approx 4.5$

YOU DO

On your own: read the worked example, then try these.

- E1 Solve $x^3 = 50$ to 1 decimal place.

- E2 Solve $x(x + 1) = 50$ to 1 decimal place.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Guess a value for the variable.
 2. Substitute it into the equation.
 3. Too big or too small? Adjust your guess.
 4. Stop when both sides are equal.

 Level 1

Is $x = 4$ a solution of $3x + 2 = 14$?

1. $3 \times 4 + 2 = 14$

Answer: yes

 Level 2

$$4m - 9 = 7$$

1. $m = 3$: 3 (low)
2. $m = 4$: 7 ✓

Answer: $m = 4$

 Level 3

$$d/3 - 4 = 2$$

1. $d = 18$: $6 - 4 = 2$ ✓

Answer: $d = 18$

 Extension

$$x^2 = 20$$

1. 4.4 low, 4.5 high

Answer: $x \approx 4.5$

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LESSON
5.11

Solving one-step equations

Date _____

Learning intentions

We are learning to:

- solve one-step equations algebraically

Success criteria

I can:

- ☐ use the inverse operation to undo $+$, $-$, $\times$ or $\div$
- ☐ do the same to both sides of the equation
- ☐ check the solution by substituting

Key terms Write the definition as your teacher explains it.

Inverse operation

Solve

Notes

Level 1 · Getting started

Easy · 20 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Solve. Show both methods.

a $w + 17 = 52$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

b $e - 12 = 0$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Solve.

14 QUESTIONS

REMEMBER Ask: what is being done to the variable?

1 $w + 17 = 52$

2 $e - 12 = 0$

3 $a + 9 = 20$

4 $m - 6 = 14$

5 $x + 25 = 40$

6 $y - 13 = 7$

Questions 7–14: solve.

7 $x + 9 = 4$

8 $b - 6 = 9$

9 $m + 4 = 4$

10 $n - 15 = 5$

11 $8p = 56$

12 $4q = -20$

13 $\frac{r}{3} = 7$

14 $c + 15 = 13$

YOU DO

On your own: **Easy** Set B. Solve.

6 QUESTIONS

1 $5p = 15$

4 $\frac{n}{5} = 9$

2 $\frac{x}{2} = 6$

5 $12c = 60$

3 $8t = 72$

6 $\frac{h}{10} = 3$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Solve. Show both methods.

a $5p = 15$

b $\frac{x}{2} = 6$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Solve. Show each step and check.

6 QUESTIONS

REMEMBER Do the **INVERSE** to both sides.

1 $-4y = 28$

3 $\frac{m}{-3} = 5$

5 $6b = 21$

2 $k + 3.5 = 1$

4 $x - 8 = -12$

6 $0.5q = 7$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve. Show both methods.

a $-4y = 28$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

b $k + 3.5 = 1$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Write an equation, then solve it.

4 QUESTIONS

REMEMBER $+$ $\leftrightarrow$ $-$, $\times$ $\leftrightarrow$ $\div$.

1 A number plus 19 is 45.

3 A number divided by 4 is 11.

2 Seven times a number is 84.

4 Take 23 from a number and the answer is 58.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO **Show off your skill**

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Solve $n + 8 = 30$.

Level 2

Solve $9x = 63$.

Level 3

Solve $\frac{t}{6} = -4$.

Extension · Stretch yourself**Worked example**

Solve $15 = x - 4$ and check.

1. The variable can be on either side
2. Add 4 to both sides: $19 = x$

Answer: $x = 19$. Check: $19 - 4 = 15$ ✓

YOU DO

On your own: read the worked example, then try these.

E1 Solve $36 = 9n$.

E2 Solve $\frac{2}{3}a = 10$.

Lesson summary

Missed the lesson? Start here.

Remember 1. Ask: what is being done to the variable?
3. $+$ $\leftrightarrow$ $-$, $\times$ $\leftrightarrow$ $\div$.

2. Do the **inverse** to both sides.
4. Check by substituting back.

Level 1

$w + 17 = 52$

1. $- 17$ from both sides

Answer: $w = 35$

Level 2

$5p = 15$

1. $\div 5$

Answer: $p = 3$

Level 3

$x/2 = 6$

1. $\times 2$

Answer: $x = 12$

Extension

$15 = x - 4$

1. $+ 4$

Answer: $x = 19$

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LESSON
5.12

Solving two-step equations

Date _____

Learning intentions

We are learning to:

- solve two-step equations algebraically

Success criteria

I can:

- ☐ undo + or – first, then $\times$ or $\div$
- ☐ set out each step on a new line
- ☐ solve equations with fractions

Key terms Write the definition as your teacher explains it.

Two-step equation

Check

Notes

Level 1 · Getting started

Easy · 18 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Solve. Undo + or – first, then $\times$ or $\div$.

a $3t + 4 = 10$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

b $4y - 7 = 5$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Questions 1–6: Solve.

12 QUESTIONS

REMEMBER Undo + or – first (both sides).

1 $2n + 6 = 12$

2 $3t + 4 = 10$

3 $5a - 2 = 18$

4 $4y - 7 = 5$

5 $6k + 1 = 31$

6 $10b - 5 = 45$

Questions 7–12: solve.

7 $2x + 3 = 11$

8 $3y - 1 = 14$

9 $5a + 2 = 22$

10 $4m - 3 = 9$

11 $\frac{k}{2} + 1 = 5$

12 $\frac{n}{3} - 2 = 4$

YOU DO

On your own: **Easy** Set B. Solve.

6 QUESTIONS

1 $\frac{m}{4} - 8 = 9$

4 $\frac{p}{5} - 3 = 1$

2 $\frac{x}{3} + 5 = 8$

5 $\frac{y}{6} + 2 = 4$

3 $\frac{a}{2} + 1 = 7$

6 $\frac{w}{10} - 1 = 0$



Level 2 · Building up

Medium · 6 questions

WE DO

With your teacher: copy what they write on the board.

Example 2

Solve equations with a fraction.

a $\frac{m}{4} - 8 = 9$

b $\frac{x}{3} + 5 = 8$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

YOU DO

On your own: **Medium** Set C. Solve. Show each step.

6 QUESTIONS

REMEMBER Then undo $\times$ or $\div$.

1 $\frac{m-4}{2} = 3$

3 $7 - 2x = 15$

5 $5d - 8 = -23$

2 $\frac{k+5}{3} = 4$

4 $3x + 11 = 2$

6 $2(x + 3) = 18$

Level 3 · Challenge

Challenging · 4 questions · then try the extension

WE DO With your teacher: copy what they write on the board.

Example 3 Solve. Show every step.

a $\frac{m-4}{2} = 3$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

b $7 - 2x = 15$

BALANCING THE SIDES

INVERSE OPERATIONS

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Solve. Show each step.

4 QUESTIONS

REMEMBER For a fraction like $(m-4)/2$: $\times$ first.

1 $\frac{2x}{3} + 1 = 9$

3 $1.5t + 2 = 8$

2 $4(y-2) = 30$

4 $\frac{3n-1}{4} = 5$

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Solve $2x + 5 = 17$.

Level 2

Solve $\frac{n}{3} - 2 = 4$.

Level 3

Solve $3(k-1) = 24$.

 Extension · Stretch yourself Worked exampleSolve $5x + 3 = 2x + 18$.

1. $- 2x$ from both sides: $3x + 3 = 18$
2. $- 3$: $3x = 15$
3. $\div 3$

Answer: $x = 5$

YOU DO

On your own: read the worked example, then try these.

E1 Solve $7a - 4 = 3a + 12$.

E2 Solve $2(x + 5) = x + 14$.

Lesson summary

Missed the lesson? Start here.

Remember

1. Undo $+$ or $-$ first (both sides).
2. Then undo $\times$ or $\div$.
3. For a fraction like $(m - 4)/2$: $\times$ first.
4. Check in the original equation.

 Level 1

$3t + 4 = 10$

1. $- 4$: $3t = 6$
2. $\div 3$

Answer: $t = 2$ Level 2

$m/4 - 8 = 9$

1. $+ 8$: $m/4 = 17$
2. $\times 4$

Answer: $m = 68$ Level 3

$(m - 4)/2 = 3$

1. $\times 2$: $m - 4 = 6$
2. $+ 4$

Answer: $m = 10$ Extension

$5x + 3 = 2x + 18$

1. $3x = 15$

Answer: $x = 5$

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
5.13

Equations and problem solving

Date _____

Learning intentions

We are learning to:

- write and solve equations to solve problems

Success criteria

I can:

- ☐ choose a variable for the unknown
- ☐ write an equation from the words
- ☐ solve it and answer the question in a sentence

Key terms Write the definition as your teacher explains it.

Unknown

Profit

Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Choose the equation, then solve it.

- a** A number multiplied by 17 is 1003. Which equation: $N + 17 = 1003$, $1003 - N = 17$, $17N = 1003$ or $1003 + N = 17$?

WRITE THE EQUATION

SOLVE IT

Your teacher works this out. Copy every step.

- b** Solve $17N = 1003$.

WRITE THE EQUATION

SOLVE IT

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: Write an equation for each. Do not solve.

10 QUESTIONS

REMEMBER Choose a letter for the unknown.

1 A number plus 8 is 20.

3 A number minus 7 is 3.

5 Double a number plus 1 is 15.

2 Five times a number is 45.

4 Half a number is 16.

6 A number times 3, minus 4, is 11.

Questions 7–10: write an equation (use n), then solve it.

7 A number plus 11 is 30.

9 A number minus 8 is 12.

8 A number times 6 is 42.

10 A number divided by 5 is 9.

YOU DO

On your own: **Easy** Set B. Solve each equation.

6 QUESTIONS

1 $n + 8 = 20$

4 $\frac{n}{2} = 16$

2 $5n = 45$

5 $2n + 1 = 15$

3 $n - 7 = 3$

6 $3n - 4 = 11$

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.**Example 2** Write an equation, then solve it.

- a** Raffle prizes cost \$180 and tickets are \$5. Write and solve an equation for the number of tickets to cover costs.

WRITE THE EQUATION

SOLVE IT

Your teacher works this out. Copy every step.

- b** How many tickets must be sold to make a \$2100 profit?

WRITE THE EQUATION

SOLVE IT

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Write an equation, solve it, and answer in a sentence.

6 QUESTIONS

REMEMBER Write an equation from the words.

- 1** A pen and a \$4 notebook cost \$6.50. Find the price of the pen.

- 3** 6 friends share a bill equally and each pays \$23. Find the bill.

- 5** A taxi charges \$4 plus \$2 per km. A trip cost \$30. How far was it?

- 2** The perimeter of a square is 52 cm. Find the side length.

- 4** Tom is 3 times as old as Kate. Tom is 36. How old is Kate?

- 6** Movie tickets cost \$15 each plus a \$3 booking fee. The total was \$78. How many tickets?

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Write an equation, solve it and answer in a sentence.

- a** The perimeter of a square is 52 cm. Find the side length.

WRITE THE EQUATION

SOLVE IT

Your teacher works this out. Copy every step.

- b** Three consecutive numbers add to 72. Find them.

WRITE THE EQUATION

SOLVE IT

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Harder problems.

4 QUESTIONS

REMEMBER Solve it step by step.

- 1** Three consecutive numbers add to 72. Find them.

- 3** Raffle prizes cost \$180 and tickets are \$5. How many tickets to make a \$2100 profit?

- 2** A rectangle is twice as long as it is wide. Its perimeter is 48 m. Find its length and width.

- 4** Ali has \$50 more than Ben. Together they have \$210. How much does Ben have?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

A phone plan costs \$20 plus 10c per text. Another costs \$35 with free texts. How many texts make them cost the same?

1. $20 + 0.1t = 35$
2. $0.1t = 15$
3. $t = 150$

Answer: 150 texts

YOU DO

On your own: read the worked example, then try these.

- E1 A gym charges \$60 to join plus \$12 a week. Another charges \$20 a week with no joining fee. After how many weeks do they cost the same?

- E2 The sum of two numbers is 50 and their difference is 14. Find the numbers.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

A number plus 14 is 31. Write and solve an equation.

Level 2

Four equal books cost \$62. Find the cost of one book.

Level 3

A number is doubled and 9 is added. The result is 41. Find the number.

Lesson summary

Missed the lesson? Start here.

Remember

1. Choose a letter for the unknown.
3. Solve it step by step.

2. Write an equation from the words.

4. Answer the question in a sentence with units.

Level 1

Number $\times 17 = 1003$

1. $17N = 1003$
2. $\div 17$

Answer: $N = 59$

Level 2

Raffle: cover \$180

1. $5N = 180$

Answer: 36 tickets

Level 3

\$2100 profit

1. $5N - 180 = 2100$
2. $5N = 2280$

Answer: 456 tickets

Extension

Phone plans

1. $20 + 0.1t = 35$

Answer: 150 texts

My reflection: what did I learn today? Which question was hardest, and why?

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